

STAT 333 Final Project

Airline Project Presentation

How to avoid cancelled flights and arrive to the destination on time during holiday seasons?

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Why Flight Reliability Matters During the Holiday Season ?

Holiday flights often face **high cancellation** and **delay rates**, especially in winter.

👉 Passengers want practical guidance on:

1.Avoiding flight cancellations

2.Maximizing on-time rate

Our project uses **BTS 2023 holiday flight data** to provide **data-driven, booking-time rules-of-thumb** for these two goals.



Rule of Thumb/Takeaways

U.S. Cancellation Advices

Avoid risky route

Choose Spirit Airline

Avoid evening departure

Travel in November instead of January

Travel on Wednesday

California → ORD

On-Time Arrival Advices

Choose Dawn flights

**Choose the airline and airports that are
historically on-time**

Avoid Tuesday

Choose November over January

1. Data Cleaning / Preprocessing

Cleaning Up the Data

- Removed rows with missing Cancelled; set Arr_Delay to NA for cancelled flights.
- Created **OnTime** indicator (1 = on time/early, 0 = late).
- Ensures clean, consistent outcomes for modeling

Fix Time Zone

- Local departure/arrival times differ across time zones.
- Converted all times to CST using a rule-based converter.
- Ensures “morning/evening” categories are comparable across all flights.

Adding Variables

- Created **Time_of_Day** categories using CST schedule times.
- Engineered booking-time predictors:
 - Route_Cancel_Rate,
 - Route_Delay_Rate, IsAirlineHub,
 - Distance_Category
- Captures historical reliability and route characteristics.

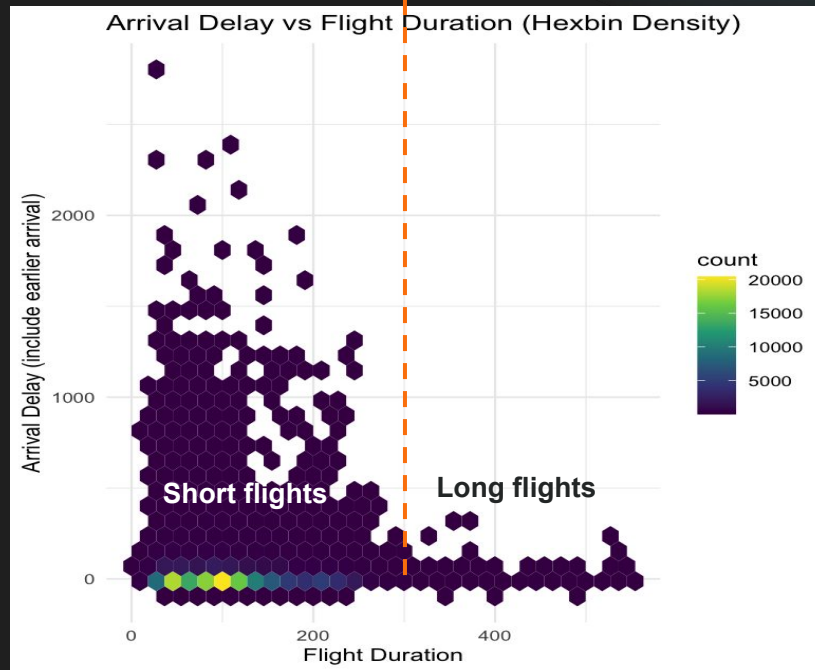
2.1 Exploratory Data Analysis (EDA):

Arrival Delay vs Flight Duration

- Long flights = more stable (delays cluster low)
- Short flights = more volatile (high delay spread)

Relationship is heteroskedastic, not linear

👉 Shorter flights are less predictable.



2.2 Exploratory Data Analysis (EDA):

Airline & Time-of-Day Delays:

1. Airline differences are large

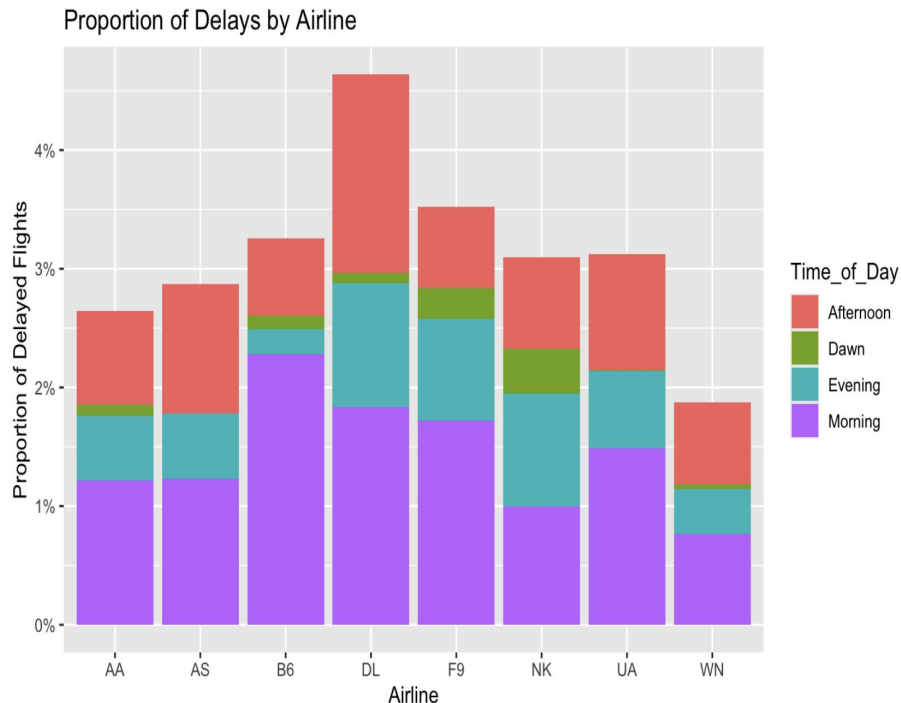
- **Southwest (WN)** - lowest delay proportion
- **Delta (DL)** - highest delay proportion

2. Time-of-day matters

- **Dawn** flights - lowest delay rate
- **Morning** flights are more delay-prone

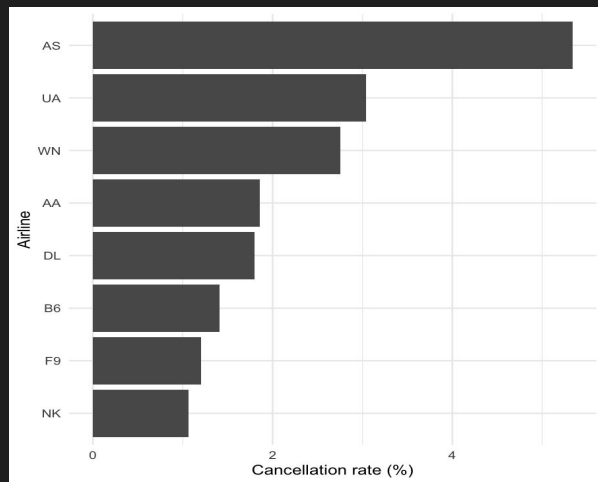
👉 This motivates including **Airline** and **Time_of_Day** as predictors in our models.

Figure 2: Proportion of delays by airlines



2.3 Exploratory Data Analysis (EDA):

Cancellation Rate by Airline

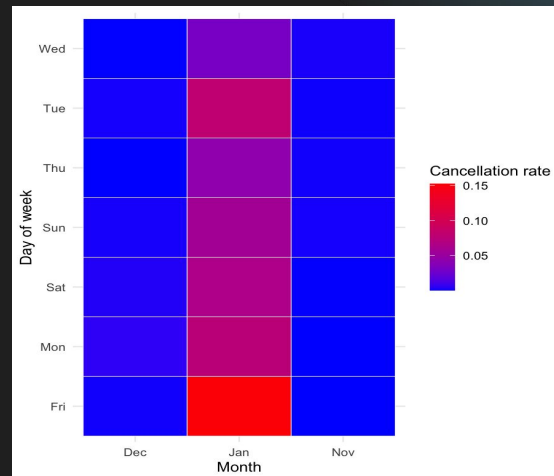


Different airlines have different cancellation rates

- **Alaska (AS)** - highest cancellation rate
- **Spirit (NK)** – lowest cancellation rate

👉 **Airline** reliability varies a lot, so Airline must be included in the model.

Cancellation Rate by Month and Weekday



Calendar effects are strong & consistent

- **January** -> much higher cancellation rates
- **Fridays** - elevated cancellation risk

👉 This justifies including Month and Weekday as predictors

2.4 Exploratory Data Analysis (EDA):

Cancellation Risk Varies Across Departure Airports

BNA & EWR :

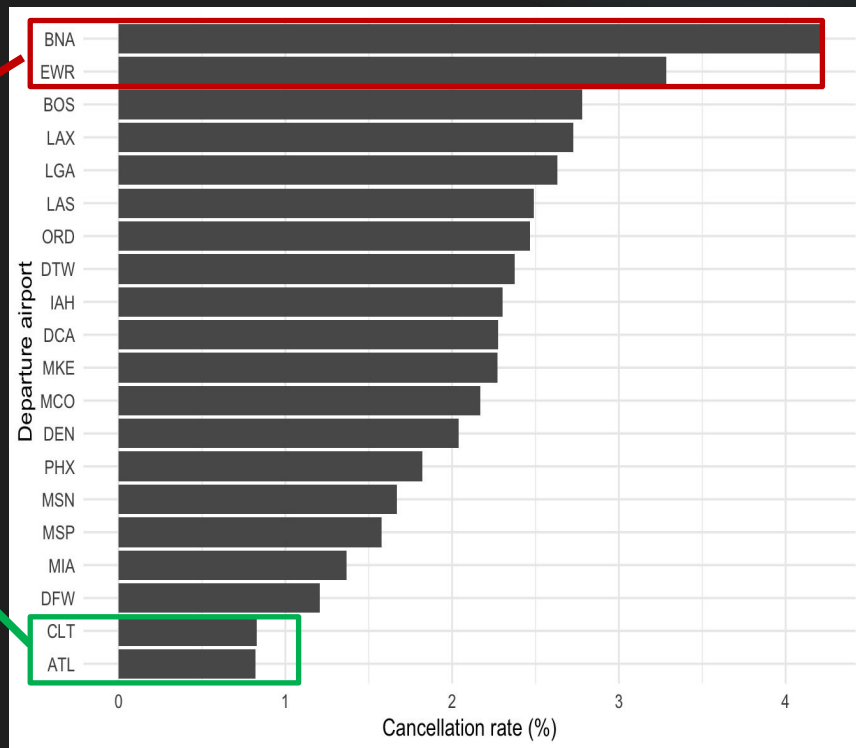
→ Top the list with consistently **higher** cancellation risk

CTL & ATL:

→ Much **lower** cancellation rates:
(likely due to better operational capacity)

👉 This suggests that the departure airport is an important predictor of **cancellation risk** and should be included in our global model.

Variation in Cancellation Risk Across Departure Airports



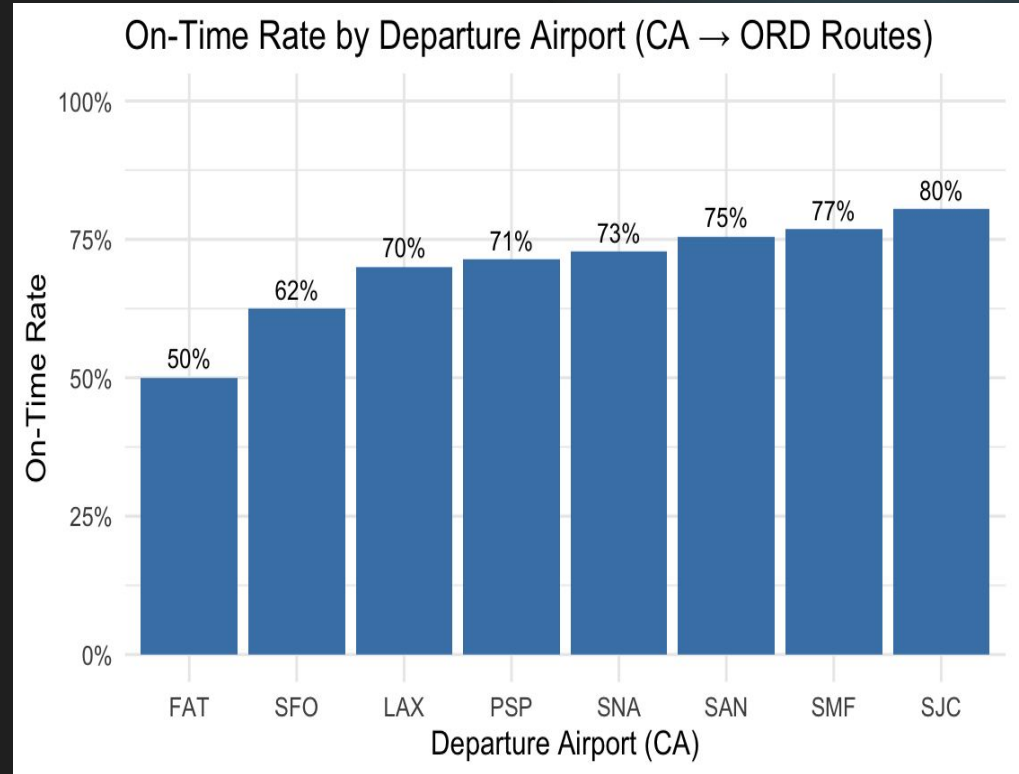
2.5 Exploratory Data Analysis (EDA):

On-time rate for different routes from CA to ORD:

- SJC, SMF, and SAN have the best performance, with about 80%, 77%, and 75% of flights on time.
- SNA, PSP, and LAX are in the middle (around 70–73% on time).
- FAT is the worst in this group with only 50% on-time.

👉 More reliable on-time performance by flying out of SJC, SMF, or SAN

👉 More cautious about flights from FAT and SFO



3. Model Choice: Logistic Regression

We modeled cancellation&ontime using logistic regression because:

Binary outcome

Both outcomes are a 0/1 variable, perfectly suited for logistic models.

Interpretability

Coefficients translate directly into practical rules (“January increases risk,” “some airlines cancel more”).

Handles categorical predictors well

Month, Week, Airline, and Airport fit naturally into the model.

Efficient for large datasets

Works reliably with our 150k+ flights.

3.1 Tips to avoid cancel flights irrespective of destination and arrival airports

What We Observed:

- Fitted a logistic regression model with outcome as cancelled flight and Airline, Dep_Airport, Month, and Time_of_Day, Route_Cancel_Rate, Week, as predictors.
- Highest Cancellation Rate in the Evening with a coefficient of 0.108435 and p-value of 0.030005.
- November has the lowest coefficient of -0.590238 with the p-value of 0.000337.
- Spirit Airlines has the lowest coefficient of -0.489035 among all airlines, with a p-value of 0.001870.
- Wednesday has the lowest coefficient of -1.702239 and a significant p-value of $< 2e-16$.
- Departure Airport ASE has the least coefficient of -1.581, p-value of 0.001196.

Conclusion/Tips:

Irrespective of the destination and arrival airports, passengers who want to minimize the risk of cancellation in our data should consider:

- Avoid historically risky route
- Carriers with lower model-estimated cancellation risk: Spirit Airline
- Avoid evening departure.
- Travel in November instead of January.
- Travel on Wednesday

Model Diagnosis

ANOVA Test

The deviance went down by 1360.1.

The p-value is $2.2e-16$, which suggests significance.

Therefore, adding variable Week and Route_Cancel_Rate gives important extra information for predicting cancellations. Model 2 is significantly better than Model 1.

AUC Test

The final cancellation model has an AUC of 0.8876.

This shows that the model can distinguish well between cancelled and non-cancelled flights. It has strong predictive ability and supports the conclusions from the model.

VIF Test

All adjusted VIF values were close to 1 (the largest was about 1.39), which is far below common cutoffs.

This means the predictors in my model are not highly correlated with each other, so multicollinearity is not a concern.

3.2 Tips for arriving on time from all airports in California to ORD airport

What We Observed:

- Fitted a logistic regression for OnTime using week_type, Dep_Airport, Airline, Time_of_Day, Elapsed_Time_Scheduled, route_ontime.
- Time of the day is a strong predictor: Coefficient of 2.009; p-value of $2.03e-17$ for dawn.
- Month is a strong predictor: Coefficient of 0.266861 and 0.088164 for November, showing significance in a 10% confidence interval
- Tuesday has a negative coefficient of -0.370315 and p-value of 0.003664.

Conclusion/Tips:

CA students who want to flying back to ORD should consider:

- Dawn flights.
- Choose the airline and airports that are historically on-time
- Avoid Tuesday
- Choose November over January

Model Diagnosis

ANOVA Test

The deviance dropped by 84.2 after adding additional predictors.

The p-value for this model is $< 4.658e-12$, which suggests significance.

Therefore, adding On_Time, Elapsed_Time_Scheduled, and interaction predictors significantly improves the fit of the model.

AUC Test

The AUC of the basic model is 0.6296, where the AUC of the improved model is 0.6559 (Train AUC 0.6556).

It suggests that our improved model is well calibrated, and reliable enough to give tips and rule of thumbs for the CA visitors or CA students at UW-Madison

VIF Test

The adjusted VIF values were all very close to 1 (the largest was about 1.82).

This means the predictors in my model are not highly correlated with each other, so multicollinearity is not a concern.

Conclusion:

For all U.S. holiday travel (Nov–Jan):

- Avoid routes with high historical cancellation rates.
- Choose carriers with lower model-estimated cancellation risk (Spirit Airlines in our data).
- Avoid evening departures.
- Travel in November instead of January when possible.
- Travel on Wednesday when schedules allow.

For CA → ORD travelers:

- Prefer dawn flights.
- Choose airlines and airports with strong historical on-time performance.
- Avoid Tuesday departures if possible.
- Travel in November rather than January when you have flexibility. If you need to travel in January, choose the morning flight is better than other options.

The results provide simple, data-driven rules of thumb that can reduce the risk of cancellations and major delays during busy holiday seasons.

Strength:

- The model uses only information that is realistically known at booking time
 - ◆ Airline
 - ◆ Departure Airport
 - ◆ Month
 - ◆ Weekday
 - ◆ Time of Day
 - ◆ Historical Route Performance
- We tried our best to avoid leakage
 - ◆ Interaction terms between Months and time of the day
 - Useful when purchase tickets.

However, there are some limitations too...

Limitations:

- ★ Single holiday period & one dataset
 - Patterns may change across years or under different operational conditions.
 - Our model cannot account for specific storms or one-off disruptions.
- ★ Limited predictors
 - We lack strong variables such as real-time weather forecasts or day-of-flight operational information.
 - The model is not intended to forecast exact on-time probabilities, but to extract stable patterns that help inform traveler decisions (for reference only).

Even though our models are not perfect, diagnostics (ANOVA, VIF, AUC) show they capture real patterns in the data.



Thank you!